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Human-Heat Interface and Its Applications in Wearable Technology

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Abstract

Heat transfer is a fundamental phenomenon that shapes our everyday life, as the human body is in a constant state of thermal exchange with its surroundings, often without our conscious awareness. Thermal homeostasis plays a critical role in regulating these exchanges, ensuring that the body maintains a stable core temperature despite fluctuations in environmental conditions. Understanding and harnessing this interaction between the human body and heat is therefore vital for advancing both health and technology. In recent years, emerging wearable electronic devices have opened new opportunities to augment natural thermoregulatory capabilities, while also enabling novel functions that extend beyond biological limits. This talk will highlight representative works on wearable human–heat interfaces, with a focus on thermal haptic devices for wearable applications, personalized thermoregulation systems for comfort and health, and energy-harvesting wearables that convert heat into usable power. Together, these advancements illustrate the future potential of integrating heat management with human-centered wearable technologies.

Keywords: ***up to 5 words***

1. Introduction

Thermal sensation is a fundamental component of human tactile perception, conveying rich information beyond simple temperature, including heat transfer dynamics and material-dependent thermal properties. During everyday interactions, humans naturally infer the thermal characteristics of objects through transient heat flow at the skin interface. However, most digital and virtual interfaces remain limited to visual and mechanical feedback, leaving thermal interaction largely unexplored.

Thermoelectric devices provide a compact and electrically controllable means of generating bidirectional heating and cooling, making them attractive for thermal-haptic interfaces. Nevertheless, conventional thermoelectric modules are typically rigid, bulky, and mechanically incompatible with the soft and deformable nature of human skin. This mismatch leads to poor conformability, reduced thermal contact, and slow or inefficient thermal response, significantly constraining their use in wearable and interactive systems.

Recent progress in soft and stretchable thermoelectric materials and device architectures has opened new opportunities for human–heat interfaces that are mechanically compliant and skin-conformal. By enabling reversible heating and cooling within a single soft device, such systems can deliver fast, localized, and repeatable thermal stimuli while maintaining performance under deformation. These characteristics are critical for applications ranging from immersive virtual environments and wearable haptics to assistive tactile interfaces and energy-related wearable systems.

In this work, we present a soft and stretchable thermoelectric platform designed as a versatile human–heat interface. The device exhibits rapid thermal response, robust operation under mechanical deformation, and the ability to reproduce material-dependent thermal sensations. Through a series of wearable and system-level demonstrations, we highlight the potential of soft thermoelectric devices as enabling components for next-generation interactive and assistive technologies.

2. Section

2.1 Subsection (Time New Roman, 9, Bold)

2.2 Scopes of MEMS conference

2.2.1. Materials, Fabrication, Packaging

2.2.1.1. Soft and Stretchable Thermoelectric Device Performance

The proposed thermoelectric platform employs a soft and stretchable device architecture that enables bidirectional heating and cooling within a single module. By incorporating elastomeric substrates and mechanically compliant interconnect designs, the device preserves stable electrical operation while accommodating large mechanical deformation. The soft structure allows intimate and conformal contact with the skin, which is essential for efficient heat transfer and reliable thermal sensation.

Thermal characterization demonstrates rapid and reversible temperature modulation under electrical actuation. The device achieves fast thermal response with characteristic response times on the order of sub-second (≈900 ms), enabling prompt transitions between heating, cooling, and neutral states. Such response speeds are sufficient to reproduce transient thermal cues experienced during real-world object interactions.

2.2.1.2. Reproduction of Material-Dependent Thermal Sensation

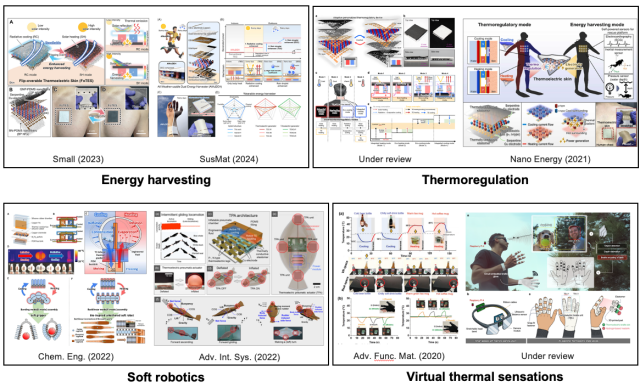
To evaluate the capability of the device to emulate realistic thermal perception, the thermoelectric actuator was programmed to reproduce thermal transients corresponding to objects with different thermal conductivities. The results show that the device can generate distinct temperature evolution profiles that qualitatively match the perceived thermal sensations of low- and

high-thermal-conductivity materials, such as paper, glass, and metal. This demonstrates that the soft thermoelectric platform can encode not only temperature but also material-dependent thermal properties through controlled heat flux modulation.

2.2.1.3. Wearable Integration and System Level Demonstrations

The mechanical compliance of the device enables seamless integration into wearable form factors, including fingertip-mounted modules and glove-based systems with motion tracking capability. In virtual reality demonstrations, the device provided synchronized thermal feedback during object contact and release, producing realistic sensations such as touching and withdrawing from cold objects. The combination of fast response and conformal skin contact significantly enhanced the realism of thermal-haptic interaction.

In addition to VR applications, the platform was extended to vision-assisted tactile interfaces, where visual information is converted into spatially distributed thermal stimuli. This demonstrates the versatility of the soft thermoelectric system as a general-purpose human–heat interface capable of supporting immersive, assistive, and interactive applications.



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